

Timelike entanglement measures for the double-scaled SYK model

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ABSTRACT: abstract

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1 Introduction

2 A spacetime density kernel for the DSSYK model

Given a Hilbert space $\mathcal{H}$ with density matrix ρ , time evolution $U_t = e^{-iHt}$, and a bipartition $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with matrix unit bases $\hat{E}_{ij}^A = |i\rangle\langle j| \otimes \mathbb{1}_B$ and $\hat{E}_{kl}^B = \mathbb{1}_A \otimes |k\rangle\langle l|$,

2.1 Definition and properties of the spacetime density kernel

For two subregions in our Hilbert space we introduce the generalized spacetime density kernel GSDK T_{AB} as the timelike entanglement equivalent to the normal density matrix ρ . The kernel encodes all time dependency of our theory and for the two-time correlators we can write

$$\text{Tr}[T_{AB}(t)(O_A \otimes O_B)] = \text{Tr}\left[\rho O_A U_t^\dagger O_B U_t\right]. \quad (2.1)$$

Furthermore it is convenient to define the components of T_{AB} which we will call the response tensor

$$T_{AB}(t) = \sum_{i,j,k,l} C_{ij;kl}(t) E_{ji}^A \otimes E_{lk}^B. \quad (2.2)$$

In this way the two-time correlator from (2.1) becomes a simple sum over the operator components

$$\mathcal{L}_t(O_A, O_B) = \text{Tr}\left[\rho O_A U_t^\dagger O_B U_t\right] = \sum_{i,j,k,l} C_{ij;kl}(t) O_{ij}^{(A)} O_{kl}^{(B)}, \quad (2.3)$$

where $O_{ij}^{(A)} = \langle i|O_A|j\rangle$ and $O_{kl}^{(B)} = \langle k|O_B|l\rangle$ are the matrix elements in the chosen basis. We calculate the response tensor as the trace

$$C_{ij;kl}(t) = \text{Tr}\left[\rho \hat{E}_{ij}^A U_t^\dagger \hat{E}_{kl}^B U_t\right]. \quad (2.4)$$

2.2 Overview of the double-scaled SYK model

The SYK model consists of N Majorana fermions which are coupled by a random parameter J in an p -body interaction.

$$H_{SYK} = i^{p/2} \sum_{i_1 \dots i_p} J_{i_1 \dots i_p} \psi_{i_1} \dots \psi_{i_p}. \quad (2.5)$$

In the double scaling limit the model is exactly solvable. We take

$$N \rightarrow \infty \quad p \rightarrow \infty, \quad (2.6)$$

while keeping

$$\lambda = \frac{2p^2}{N} = \text{const.} \quad (2.7)$$

constant. We also define the parameter $q = e^{-\lambda}$, which is used as the deformation parameter in the later part.

In this limit the moments of the Hamiltonian

$$m_k = \left\langle \text{Tr} H^k \right\rangle_J \quad (2.8)$$

can be calculated as a sum over chord diagrams as showed in e.g. [1, 2]. By evaluating this sum combinatorically we go to an auxiliary Hilbert space of open chord states. With states $|l\rangle$ as the basis of the chord number states we can define the transfer matrix

$$T = \begin{pmatrix} 0 & 1 & 0 & \dots \\ 1 & 0 & & \end{pmatrix} \quad (2.9)$$

2.3 Derivation of the DSSYK spacetime density kernel

$$C_{ij;kl}(t) = \langle 0|\rho|i\rangle \int \frac{d\theta d\phi}{(2\pi)^2} \mu(\theta)\mu(\phi) e^{-it(E(\theta)-E(\phi))} \frac{H_j(\cos\theta|q)H_k(\cos\phi|q)}{\sqrt{(q;q)_j(q;q)_k}} \frac{H_l(\cos\theta|q)}{\sqrt{(q;q)_l}}$$

2.4 Relation to the two-point function

3 Numerical results

3.1 Timelike entanglement measures

The spacetime density kernel allows us to study correlations in time, since it encodes the dynamics of different time-slices in one operator. Multiple measures, which quantify these correlations, can be extracted from the spacetime density kernel $T_{AB}(t)$. The goal of this section is to numerically calculate these quantities, which were introduced in [3], for the double-scaled SYK model. It is important to note that $T_{AB}(t)$ is not necessarily Hermitian, but $\text{Tr } T_{AB}(t) = 1$ always holds. Firstly, we aim to calculate the Tsallis entropy

$$Z_n(t) = \text{Tr}[T_{AB}(t)^n], \quad (3.1)$$

while we mostly focus on the $n = 2$ case. We also investigate the Schatten p -norms of T_{AB} , which are defined by

$$\|M\|_p = \text{Tr} \left((M^\dagger M)^{p/2} \right)^{1/p}. \quad (3.2)$$

In particular, we are interested in the Frobenius norm

$$\|T_{AB}(t)\|_2^2 = \text{Tr} \left[T_{AB}(t) T_{AB}^\dagger(t) \right]. \quad (3.3)$$

In order to quantify the non-Hermiticity of the spacetime density kernel, we also study the entanglement p -magitivity

$$\mathcal{I}_p(t) = \|T_{AB}(t) - T_{AB}^\dagger(t)\|_p. \quad (3.4)$$

Here, we focus again on the case $p = 2$, since $\mathcal{I}_2(t)$ controls causal influence: it is nonzero iff signaling between A and B is possible.

Another important quantity which we will consider is the kernel negativity $\mathcal{N}_H(t)$, which quantifies the spectral weight of the Hermitian part $H_{AB}(t) \equiv (T_{AB} + T_{AB}^\dagger)/2$ of the spacetime density kernel. It is defined by

$$\mathcal{N}_H(t) = \sum_{\mu_\alpha(t) < 0} |\mu_\alpha(t)|, \quad (3.5)$$

where $\{\mu_\alpha(t)\}$ are the eigenvalues of $H_{AB}(t)$.

3.2 Spectral form factor

References

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